

Unit 3: Network Layer Services & IPv4 Classless Addressing

Handwritten Student Notebook

Syllabus Topics Covered:

OFFICIAL SYLLABUS TOPICS COVERED Network Layer Services: Packetizing, Forwarding and Routing, Error control, Flow control, Congestion control. Packet Switching Models: Datagram approach (Connectionless) vs Virtual-Circuit approach (Connection-oriented). Network ...

Table of Contents

Network Layer Services & Switching Models	p. 2
Network Performance & Delay Modeling	p. 3
IPv4 Addressing & CIDR Subnetting	p. 5
VLSM Hierarchical Subnet Design	p. 9
One-Page Rapid Revision Sheet (CIE-1 Scope)	p. 10
Important Examination Questions (CIE-1 Scope)	p. 11
Practice Problems with Step-by-Step Solutions (CIE-1 Scope)	p. 13

Network Layer Services & Switching Models

Source: Forouzan Se, Ch. 18.1–18.2, pp. 521–528; ISS3-CN-Unit3 slides 1–25

§ 1

The **Network Layer** is responsible for host-to-host delivery of individual packets across multiple diverse network links. It encapsulates transport-layer segments into network **datagrams** at the sender, routes them across intermediate routers, and delivers them to the destination host.

THREE FUNDAMENTAL NETWORK LAYER SERVICES:

1. **Packetizing:** Encapsulating the transport-layer payload into a network-layer datagram by adding an IP header at the source, and decapsulating it at the destination.
2. **Routing:** The global decision-making process of determining the optimal, end-to-end path that packets should traverse from source to destination using routing protocols (OSPF, BGP, RIP).
3. **Forwarding:** The local action taken by each individual router of transferring a packet from an incoming interface link to the appropriate outgoing interface link based on its forwarding table.

Datagram (Connectionless) vs Virtual-Circuit (Connection-Oriented) Networks

Feature / Dimension	Datagram Approach (e.g. The Internet / IP)	Virtual-Circuit Approach (e.g. ATM, X.25)
Setup Phase	None. Packets can be injected immediately.	Mandatory connection setup phase before sending data; teardown after.
Packet Header	Contains full 32-bit (or 128-bit) Source and Destination IP addresses.	Contains only a short Virtual Circuit Identifier (VCI) label.
Routing Path	Each datagram is routed independently. Packets between same source and destination may take completely different paths.	All packets for a connection follow the exact same established physical path.

Feature / Dimension	Datagram Approach (e.g. The Internet / IP)	Virtual-Circuit Approach (e.g. ATM, X.25)
Packet Ordering	Packets may arrive out-of-order at destination; receiver must resequence.	Packets guaranteed to arrive strictly in the order sent.
Router State & Robustness	Routers maintain no per-connection state . Highly robust against router crashes (traffic automatically reroutes).	Routers maintain connection state tables. If any intermediate switch crashes, the entire connection is severed.

QUICK RECALL

- Routing = global path selection (control plane); Forwarding = local packet switching (data plane).
- The Internet uses the **Datagram model**: connectionless, independent packet routing, robust against router failure.
- Virtual-circuit uses setup/teardown phases and VCI labels.

Network Performance & Delay Modeling

Source: Forouzan Se, Ch. 3.6 & 18.3, pp. 95–98, 529–530; IS53-CN-Unit3 slides 26–38

§ 2

As a datagram journeys from source host to destination, it encounters four distinct delay components at every intermediate router. Understanding these delays is essential for network dimensioning and quality of service (QoS).

TOTAL NODAL DELAY EQUATION:

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

1. **Processing Delay (d_{proc}):** The time required for a router to inspect the packet header, check bit-level checksums, and look up the destination IP in the routing table (microseconds).
2. **Queuing Delay (d_{queue}):** The time a packet spends waiting in the router's transmission buffer queue before being placed on the link. Highly variable, depends on traffic congestion and queue depth.
3. **Transmission Delay (d_{trans}):** The time required to push all bits of the packet onto the physical transmission link:

$$d_{\text{trans}} = \frac{L}{R} \quad [\text{seconds}]$$

where L is packet length (bits) and R is link transmission rate / bandwidth (bps).

4. **Propagation Delay (d_{prop}):** The time required for a single bit to travel physically across the transmission medium from source to destination:

$$d_{\text{prop}} = \frac{d}{v} \quad [\text{seconds}]$$

where d is distance (meters) and v is propagation velocity of the medium ($\approx 2 \times 10^8$ m/s in copper/fiber, 3×10^8 m/s in air).

CRITICAL DISTINCTION: TRANSMISSION VS PROPAGATION DELAY:

Transmission delay (L/R) depends on packet size and transmission rate (bandwidth). It is the time to push the bits into the pipe.

Propagation delay (d/v) depends solely on physical distance and propagation velocity. It is the time for a bit to travel through the pipe.

Bandwidth-Delay Product (BDP)

The **Bandwidth-Delay Product** defines the volume of bits that can fill the transmission link end-to-end at any instant (the "pipe volume"):

$$\text{BDP} = \text{Bandwidth} \times \text{Delay} = R \times t_{\text{prop}}$$

Example: A 100 Mbps link with a 20 ms one-way propagation delay has a BDP of $(100 \times 10^6 \text{ bps}) \times (0.020 \text{ s}) = 2,000,000 \text{ bits} = 250 \text{ KB}$. The sender can emit 250 KB before the first bit reaches the receiver!

QUICK RECALL

- $d_{\text{trans}} = L/R$ (packet length / bandwidth); $d_{\text{prop}} = d/v$ (distance / speed).
- Total delay = $d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$.
- $\text{BDP} = R \times t_{\text{prop}}$ represents maximum bits in flight.

IPv4 Addressing & CIDR Subnetting

Source: Forouzan Se, Ch. 18.4, pp. 531–545; IS53-CN-Unit3 slides 39–70

§ 3

An **IPv4 address** is a 32-bit binary identifier uniquely assigned to every host network interface card connected to the global Internet. Addresses are conventionally written in **dotted-decimal notation** (e.g. '192.168.1.1'), where each 8-bit octet is represented as a decimal number between 0 and 255.

1. Historic Classful Addressing Architecture

Class	Leading Bits	First Octet Range	Default Mask	NetID / HostID Split	Number of Networks & Hosts
Class A	'0...'	'1.0.0.0' – '126.255.255.255'	'255.0.0.0' (18)	8 bits NetID, 24 bits HostID	126 networks, $2^{24} - 2 = 16,777,214$ hosts each
Class B	'10...'	'128.0.0.0' – '191.255.255.255'	'255.255.0.0' (16)	16 bits NetID, 16 bits HostID	16,384 networks, $2^{16} - 2 = 65,534$ hosts each

Class	Leading Bits	First Octet Range	Default Mask	NetID / HostID Split	Number of Networks & Hosts
					2,097,152 networks, $2^8 - 2 = 254$ hosts each
Class C	<code>'110...'</code>	<code>'192.0.0.0' - '223.255.255.255'</code>	<code>'255.255.255.0' (/24)</code>	24 bits NetID, 8 bits HostID	
Class D	<code>'1110...'</code>	<code>'224.0.0.0' - '239.255.255.255'</code>	N/A	Reserved for Multicast	No NetID/HostID distinction
Class E	<code>'1111...'</code>	<code>'240.0.0.0' - '255.255.255.255'</code>	N/A	Reserved for Experimental	Future research and testing

Special Note: `'127.0.0.0/8'` is reserved for loopback testing (`127.0.0.1` = localhost).

2. Classless Inter-Domain Routing (CIDR)

Classful addressing caused massive address wastage (a company needing 500 hosts had to take a Class B with 65,534 addresses). In 1993, **CIDR (Classless Inter-Domain Routing)** abandoned fixed classes in favor of arbitrary prefix lengths specified using **slash notation (/n)**.

CIDR BLOCK MATHEMATICAL LAWS:

For an IP block specified as **A.B.C.D/n**:

1. **Prefix length (n)**: Number of 1s in the subnet mask ($0 \leq n \leq 32$).
2. **Host bits (h)**: $h = 32 - n$.
3. **Total IP addresses in block (N)**:

$$N = 2^{32-n} = 2^h$$

4. **Total assignable usable hosts (N_{usable})**:

$$N_{usable} = 2^{32-n} - 2$$

(Subtract 2: one for Network Address, one for Directed Broadcast Address).

5. **Network Address**: Bitwise AND between IP address and subnet mask.
6. **Directed Broadcast Address**: Bitwise OR between IP address and inverted mask (replace all h host bits with 1s).
7. **First Usable Host IP**: Network Address + 1.
8. **Last Usable Host IP**: Broadcast Address - 1.

WORKED NUMERICAL · VERIFIED SCRIPT PASS [8 MARKS · EXAM FAVOURITE]

Comprehensive Analysis of CIDR Block: 200.16.70.82/27

Given the host IPv4 address **200.16.70.82/27**, determine:

1. Subnet mask in dotted-decimal format
2. Network address (Subnet ID)
3. Directed broadcast address
4. Range of usable host IP addresses
5. Total number of assignable host addresses

Step 1: Subnet Mask:

Prefix length $n = 27$ means twenty-seven 1s followed by five 0s:

$$\text{Mask} = \underbrace{11111111}_{255} . \underbrace{11111111}_{255} . \underbrace{11111111}_{255} . \underbrace{11100000}_{224} = \mathbf{255.255.255.224}$$

Step 2: Network Address:

Focus on the 4th octet: $82 = 01010010_2$. Mask 4th octet = 11100000_2 .

$$01010010_2 (82)$$

$$\text{AND } 11100000_2 (224) = 01000000_2 = 64$$

$$\text{Network Address} = \mathbf{200.16.70.64}$$

Step 3: Directed Broadcast Address:

Set all $32 - 27 = 5$ host bits in 4th octet to 1s:

$$01000000_2 \text{ OR } 00011111_2 = 01011111_2 = 64 + 31 = 95$$

$$\text{Broadcast Address} = \mathbf{200.16.70.95}$$

Step 4: Usable Host Range & Usable Count:

- First usable host: Network Address + 1 = **200.16.70.65**
- Last usable host: Broadcast Address - 1 = **200.16.70.94**
- Total usable hosts: $2^{32-27} - 2 = 2^5 - 2 = 32 - 2 =$
30 usable hosts.

Example CIDR Block: 200.16.70.82 / 27 (Subnet Mask: 255.255.255.224)
Total 32 Bits = 27 Network Prefix Bits + 5 Host Identifier Bit



Figure 3.2: CIDR Prefix Bit Partition & Host Range for 200.16.70.82/27

As required in Ramaiah SEE Examination: 27 network prefix bits leave 5 host bits, yielding 30 assignable host addresses (200.16.70.65 to 200.16.70.94).

[Interactive Visualizer Online]: Interactive IPv4 Subnet & CIDR Calculator. Explore the step-by-step visualizer on the live course notes page.

QUICK RECALL

- Class A = /8, Class B = /16, Class C = /24. Class D = Multicast.
- CIDR $/n$: $h = 32 - n$. Total IPs = 2^h ; Usable hosts = $2^h - 2$.
- Network address has all host bits = 0; Broadcast address has all host bits = 1.

VLSM Hierarchical Subnet Design

Source: Forouzan Se, Ch. 18.4, pp. 546–555; IS53-CN-Unit3 slides 71–85

§ 4

Variable Length Subnet Masking (VLSM) enables network engineers to allocate subnets of different sizes tailored precisely to the number of hosts required in each departmental segment, completely eliminating address fragmentation.

Dept	Hosts Needed	Formula ($2^h - 2 \geq$ Needed)	Allocated Block	Subnet Mask	Usable Host IP Range
		$2^7 - 2 =$			
Dept A	120 hosts	$126 \geq 120 \Rightarrow h = 7, n = 25$	'192.168.1.0/25'	'255.255.255.128'	'192.168.1.1' - '192.168.1.126'
		$2^6 - 2 =$			
Dept B	60 hosts	$62 \geq 60 \Rightarrow h = 6, n = 26$	'192.168.1.128/26'	'255.255.255.192'	'192.168.1.129' - '192.168.1.190'
		$2^5 - 2 =$			
Dept C	30 hosts	$30 \geq 30 \Rightarrow h = 5, n = 27$	'192.168.1.192/27'	'255.255.255.224'	'192.168.1.193' - '192.168.1.222'
		$2^4 - 2 =$			
Dept D	10 hosts	$14 \geq 10 \Rightarrow h = 4, n = 28$	'192.168.1.224/28'	'255.255.255.240'	'192.168.1.225' - '192.168.1.238'

QUICK RECALL · VLSM DESIGN

- **VLSM Golden Rule:** Always sort departmental host requirements in strictly descending order before allocating blocks.
- **Host formula:** $2^h - 2 \geq \text{extrequiredhosts}$ to find the number of host bits h .
- **Subnet prefix:** New prefix $n = 32 - h$.
- **Block alignment:** Subnet base address must be an integer multiple of the block size 2^h .

One-Page Rapid Revision Sheet (CIE-1 Scope)

REVISION

Network Delays

- $d_{trans} = L/R$ (packet length / bandwidth).
- $d_{prop} = d/v$ (distance / propagation speed).
- Total nodal delay: $d_{proc} + d_{queue} + d_{trans} + d_{prop}$.
- $extBDP = R \cdot d_{prop}$ ("pipe volume in bits").

CIDR & VLSM Formulas

- Prefix `/n` has $h = 32 - n$ host bits.
- Total addresses = 2^h ; Usable hosts = $2^h - 2$.
- Network IP: AND with mask. Broadcast IP: OR with inverted mask.
- VLSM: always sort subnets by host count descending!

Switching Models

- **Datagram:** Connectionless, no setup, independent routing per packet, robust to node failures.
- **Virtual Circuit:** Connection-oriented, 3-phase (setup, transfer, teardown), VC table lookup, in-order delivery.

CIDR Block Restrictions

- **Contiguity:** Addresses in a block must be consecutive numbers.
- **Power of 2:** Total addresses in a block must be 2^{32-n} .
- **Divisibility:** Starting IP must be evenly divisible by block size.

Important Examination Questions (CIE-1 Scope)

? QUESTIONS

2-Mark Questions (Short Answer)

Practice Problem:

2M Distinguish between routing and forwarding.

Solution:

Answer: Routing is the global process of determining the end-to-end optimal path across the entire network (control plane). Forwarding is the local operation at a router transferring a packet from its input link to the appropriate output link based on its forwarding table (data plane).

Practice Problem:

2M What is the Bandwidth-Delay Product (BDP)? What does it physically represent?

Solution:

Answer: BDP is the product of a channel's bandwidth (in bps) and its one-way propagation delay (in seconds). It represents the maximum volume of bits that can fill the transmission pipe at any instant.

Practice Problem:

2M State the three restrictions imposed on classless (CIDR) address blocks.

Solution:

Answer: (1) Addresses in the block must be contiguous. (2) Total number of addresses must be a power of 2 (2^{32-n}). (3) First address must be evenly divisible by the block size.

5-Mark & 10-Mark Questions

Practice Problem:

6M Compare Datagram (connectionless) vs Virtual Circuit (connection-oriented) network service.

Solution:

Answer: Refer to Section 1 comparison table. Key points: Connection setup phase, packet addressing (full IP vs VCI label), routing path (independent dynamic vs fixed path), router state requirements, and robustness under node failure.

Practice Problem:

10M Design an address allocation for three customer groups from 120.50.0.0/16.

Solution:

Answer: See full solved problem in [PYQ Answer Bank \(Q17\)](#) with exact prefix calculations and address boundary table.

Practice Problems with Step-by-Step Solutions (CIE-1 Scope)

PRACTICE

Test your understanding with these exam-level numericals. Click each collapsible card to reveal the complete verified solution:

PROBLEM 1 · CIDR ADDRESSING

An organization is assigned the CIDR block **198.51.100.160/26**. Determine: (a) Subnet Mask in dotted decimal, (b) Network Address, (c) Broadcast Address, and (d) Range of usable host IP addresses.

Practice Problem:

Show Step-by-Step Solution

Solution:**Step 1: Subnet Mask**

Prefix length $n = 26$. First 3 octets are all 1s (24 bits). 4th octet has 2 ones ($11000000_2 = 192_{10}$).

$$\text{Subnet Mask} = \mathbf{255.255.255.192}$$

Step 2: Network Address

Bitwise AND on 4th octet: $160_{10} = 10100000_2$.

$$10100000_2 \text{ AND } 11000000_2 = 10000000_2 = 128_{10}.$$

$$\text{Network Address} = \mathbf{198.51.100.128}$$

Step 3: Broadcast Address

Host bits $h = 32 - 26 = 6$. Total addresses in block $= 2^6 = 64$.

$$\text{Broadcast} = 128 + 64 - 1 = 191.$$

$$\text{Broadcast Address} = \mathbf{198.51.100.191}$$

Step 4: Usable Host Range

$$\text{Usable Hosts} = \mathbf{198.51.100.129 \text{ to } 198.51.100.190} \quad (\text{Total } 62 \text{ hosts})$$

PROBLEM 2 · VLSM ALLOCATION

An enterprise base block is **192.168.10.0/24**. Allocate subnets for three departments: Dept A (60 hosts), Dept B (28 hosts), and Dept C (12 hosts). Specify network address, mask, and host range for each.

Practice Problem:

Show Step-by-Step Solution

Solution:

Always allocate starting from the largest requirement:

- **Dept A (60 hosts):** Needs $60 + 2 = 62$ addresses. Nearest power of 2 is $2^6 = 64$ ($h = 6$). Subnet prefix: $32 - 6 = /26$ (Mask 255.255.255.192).
 - Network: 192.168.10.0/26, Range: 192.168.10.1 - 192.168.10.62, Broadcast: 192.168.10.63.
- **Dept B (28 hosts):** Needs $28 + 2 = 30$ addresses. Nearest power of 2 is $2^5 = 32$ ($h = 5$). Subnet prefix: $32 - 5 = /27$ (Mask 255.255.255.224).
 - Starts at next free boundary: 192.168.10.64/27, Range: 192.168.10.65 - 192.168.10.94, Broadcast: 192.168.10.95.
- **Dept C (12 hosts):** Needs $12 + 2 = 14$ addresses. Nearest power of 2 is $2^4 = 16$ ($h = 4$). Subnet prefix: $32 - 4 = /28$ (Mask 255.255.255.240).
 - Starts at next free boundary: 192.168.10.96/28, Range: 192.168.10.97 - 192.168.10.110, Broadcast: 192.168.10.111.

PROBLEM 3 · TRANSMISSION VS PROPAGATION DELAY

A 1500-byte packet is sent over a 2500 km fiber-optic cable ($v = 2 \times 10^8$ m/s) at transmission rate $R = 1$ Gbps. Calculate d_{trans} , d_{prop} , and determine whether the link is latency-dominated or bandwidth-dominated.

Practice Problem:

Show Step-by-Step Solution

Solution:**1. Transmission Delay:**

$$L = 1500 \times 8 = 12,000 \text{ bits}$$

$$d_{trans} = \frac{L}{R} = \frac{12,000 \text{ bits}}{10^9 \text{ bps}} = 1.2 \times 10^{-5} \text{ s} = \mathbf{12 \mu s}$$

2. Propagation Delay:

$$d = 2,500 \text{ km} = 2.5 \times 10^6 \text{ m}$$

$$d_{prop} = \frac{d}{v} = \frac{2.5 \times 10^6}{2 \times 10^8} = 0.0125 \text{ s} = \mathbf{12.5 \text{ ms} = 12,500 \mu s}$$

3. Latency Regime:

$$d_{prop}/d_{trans} = 12,500/12 \approx \mathbf{1041.7}$$

Since propagation delay exceeds transmission delay by over 1000 $\times$, this long-haul link is heavily **propagation/latency-dominated**.